Morphological embedding factorization helps language models in proportion to a language's morphology

Draft — internal. Single machine, treebank/subset scale, 2–3 seeds.
Read §7 before citing any figure.

Abstract

We test whether an explicit morphological prior—factoring a word's input embedding into a lemma embedding plus an operator for its inflection—improves next-token prediction, measured in bits/char of the original text (the only tokenizer-independent measure). In English the effect is negligible (-0.011 bits/char). Across five morphologically richer languages it is 20–40× larger (-0.19 to -0.42 bits/char), scaling with how much of the language is inflected. A shuffled-root control shows the gain is largely due to the *low-rank pooling structure* of tying rare word-forms to shared roots, not to the *correctness* of the morphological grouping—the morphology-specific component is clearly present only in Turkish. Derivation contributes nothing for a language model in any functional form we tried, and "operator as a token" is strictly worse than "operator as a pre-model affine map." The paper doubles as a catalogue of measurement traps: nearly every strong effect we found was, on first measurement, a confound in the direction of our hopes.

1 Introduction

Language models tokenize with subword schemes—byte-pair encoding and its relatives—that are optimized for compression, not morphology. A word like Finnish *taloissamme* ("in our houses") is shattered into pieces that cut across morpheme boundaries, and every inflected form of a lemma is estimated largely independently. This is wasteful exactly where a language inflects heavily: the model must learn each of a lemma's dozens or hundreds of surface forms as a separate distribution.

We ask a narrow, quantitative question: if we hand the model an explicit morphological prior —factoring a word's input embedding into a lemma embedding plus a learned operator for its inflection—does next-token prediction improve, by how much, and *why*? We measure in bits/char of the original text, the only measure comparable across tokenizers, and we control the result three ways: against a byte-fallback BPE baseline that cannot cheat by discarding rare words; against a pure word-level tokenizer, to separate the tokenizer from the factorization; and against a shuffled-root null that keeps the factorization's capacity but destroys the *correctness* of the morphological grouping.

Our contributions are (i) a controlled cross-linguistic measurement showing the prior helps in proportion to a language's inflection, negligibly in English and 20–40× more in five richer languages; (ii) the finding, from the shuffled-root null, that most of the gain is generic low-rank pooling rather than morphology-specific alignment; (iii) the negative results that derivation contributes nothing to a language model and that operator-as-token is worse than operator-as-affine-map; and (iv) a catalogue of measurement traps, since nearly every effect we report was, on first measurement, a confound in the direction of our hypothesis.

2 Related work

(Bibliography is a TODO; the works below are named for orientation and their exact citations should be verified before circulation.)

Subword tokenization. BPE (Sennrich et al.) and unigram/SentencePiece (Kudo) are the standard; subword regularization and BPE-dropout (Kudo; Provilkov et al.) show that tokenization is a modeling choice with real perplexity consequences. Unsupervised morphological segmentation (Morfessor; Creutz & Lagus) predates the neural era. Our baseline is byte-level BPE with a byte fallback, and our central metric point is that the apparent sequence-length cost of word-level tokenization is mostly an `<unk>`-accounting artifact (§6).

Morphology-aware and factored embeddings. Character-aware neural language models (Kim et al.) and compositional-morphology embeddings (Botha & Blunsom; Vania & Lopez) build word representations from sub-lexical structure. Our factorization is deliberately minimal—a lemma embedding plus a low-rank affine operator per inflection slot, precomputed into an ordinary embedding table—so it is a structured prior at zero inference cost rather than a runtime character encoder. The novel component is the *shuffled-root null*, which lets us ask not whether structure helps but whether the *morphological* structure specifically helps.

Multilingual and low-resource morphology. Massively multilingual models (mBERT, XLM-R) and the Universal Dependencies treebanks make controlled cross-linguistic measurement possible; we use UD's gold lemma-and-features as the factorization, avoiding an analyzer confound. Our practical conclusion—that the prior pays most in the data-scarce, morphologically-rich regime—points at low-resource agglutinative and fusional languages as the setting where it matters.

3 Setup

Metric. bits/char over the original characters, counting only characters inside the scored window. Cross-entropy is *not* comparable across tokenizers (a lexeme stream contains low-entropy operator tokens), so we never compare it. **Baseline.** GPT-2 byte-level BPE, 16k vocab, with a byte fallback so no token is ever `<unk>` (§6). **Trunk.** A small GPT (L6, d=384; 10.65M non-embedding params). Embedding parameters are excluded from "model size": they are a lookup table, zero FLOPs.

The factorization. A word's embedding is

$$e(\text{word}) = E_{\text{root}} + d_{\text{slot}} + U_{\text{slot}}\,(V_{\text{slot}}^{\top} E_{\text{root}}), \qquad \text{rank } 32, \tag{1}$$

with E_{root} the lemma embedding, d_{slot} a per-inflection translation, and $UV^{\top}$ a low-rank rotation. The identity slot is pinned (d_0=0, U_0=V_0=0) to remove a gauge freedom that otherwise makes the operator parameters unidentifiable. This precomputes into an ordinary $[V, d]$ table: **zero inference FLOPs**. It is a structured prior on the embedding matrix, not a runtime component.

4 English: the prior barely helps, and derivation not at all

On wikitext-103, inflection-only factoring (7 operators) beats BPE, and the advantage *grows as data shrinks* (Table 1, Fig. 3). Each arm is trained to its own best held-out checkpoint under a

Table 1: English data-efficiency ladder (wikitext-103, byte fallback, best held-out checkpoint, 2 seeds).

paragraphs	BPE	inflection-factored	Δ
10,000	1.956	1.824	−0.132
40,000	1.771	1.689	−0.082
160,000	1.740	1.665	−0.075

byte fallback; at the small rung both arms genuinely converge, so the gain is a fair best-vs-best comparison—though at 10k it is partly *overfitting resistance*, a legitimate sample-efficiency benefit.

Derivation contributes nothing. Adding derivational operators (*worker* ← *work*, …)—as tokens or affine maps, with rotation allowed, at zero sequence-length cost, composed to depth 4—never beats inflection-only ($\Delta = -0.001 \pm 0.003$). Seven inflectional operators do all the work; the 27 derivational operators and the entire 29-relation "atlas" buy zero bits/char for a language model. (The atlas's *geometric* claims about embedding space survive their controls; what dies is the inference from "English has this structure" to "an LM benefits from it.")

Operator as matrix, not token. A pre-model affine map beats an operator token and beats a free embedding, and survives its shuffled-root null. A pure translation ($E_{\text{root}} + d_{\text{slot}}$, no rotation) *loses* to a free embedding; the rotation term is necessary. This contradicts the atlas's "inflection is a translation" prediction *inside* a language model: the bias is necessary but not sufficient.

Fig. 1 ranks what drives the English gain: decomposing many *rare* words costs tokens and buys nothing; the win is decomposing *frequent* forms; false decompositions matter an order of magnitude less.

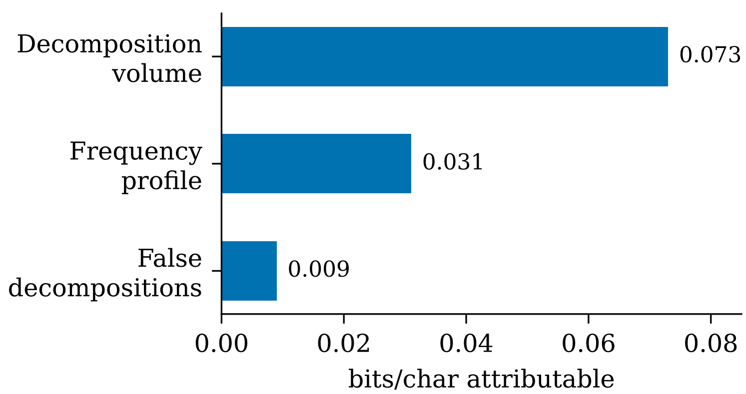

Figure 1: Drivers of the English gain. Volume (decomposing rare words) dominates and is a *cost*, not a benefit; the payoff is in the frequency profile.

5 Cross-linguistic: the prior scales with morphology

Universal Dependencies treebanks (gold lemma + features = root + operator; no analyzer or sandhi confound), 3 seeds. *affine* − *free* is the factorization gain; *affine* − *shufroot* (random-root pointing, matched capacity) isolates the morphology-specific component (Table 2, Fig. 2).

Finding 1 (robust). The factorization gain is 20–40× larger in every rich language than in English; the ordering is roughly monotonic in inflection rate. English is close to the *worst* language to demonstrate this on.

Table 2: Cross-linguistic factorization gain. *affine–free* is the total gain; *affine–shufroot* is the morphology-specific part (random-root null).

language	% content-word inflected	affine − free	affine − shufroot
English	30	−0.011	−0.007
Spanish	40	−0.189	−0.017
German	41	−0.216	+0.012
Turkish	66	−0.359	−0.044
Russian	71	−0.419	−0.016
Finnish	80	−0.229	−0.005

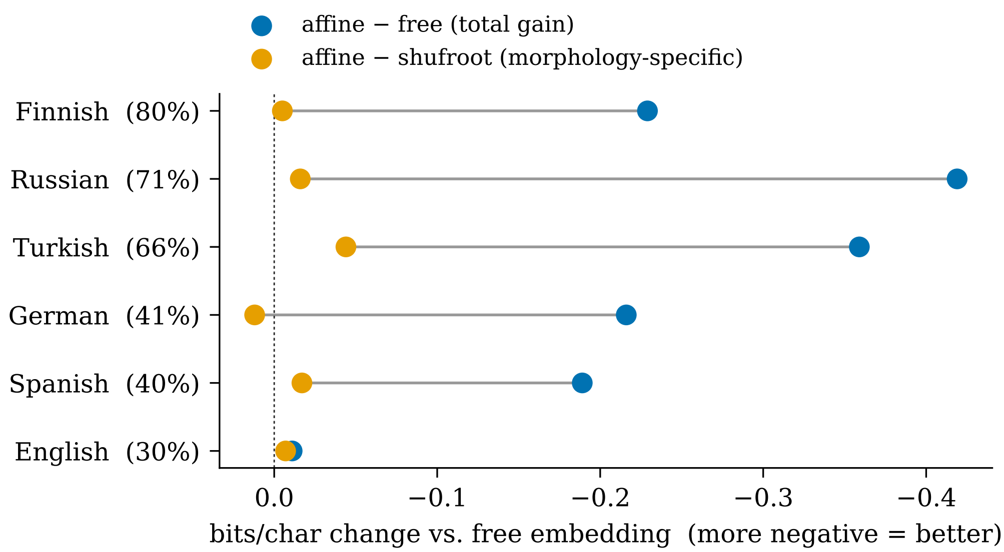

Figure 2: The total factorization gain (blue) is large in every morphologically rich language and negligible in English; the morphology-*specific* component (orange, vs. a shuffled-root null) is near zero except in Turkish. Languages ordered by inflection rate.

Finding 1a (the Finnish outlier is corpus size, not morphology). Finnish is a low outlier (−0.229 at 80% inflection, below Turkish's −0.359 at 66%) only because its treebank is 2.3× larger. The gain is a sample-complexity effect (cf. the English ladder, Fig. 3): halving the Finnish corpus grows the gain to −0.391, *above* Turkish, restoring monotonicity. The cross-linguistic table therefore conflates morphological richness with corpus size—a confound (see §7)—but the resolution reinforces the core claim: less data, bigger gain, in every language.

Finding 2 (the honest caveat). The morphology-specific component is clearly nonzero only in Turkish; in German it is even slightly positive (shuffled beats real). Most of the gain is the low-rank pooling structure of tying sparse word-forms to shared roots, which helps regardless of whether the grouping is the *correct* morphology. We therefore frame this as *structured sparse embedding factorization helps morphologically rich languages*, not "morphology is the mechanism."

Finding 3. Word-level tokenization *alone* loses to BPE in sparse agglutinative languages (Turkish +0.155, Russian +0.405, Finnish +0.218) because word-forms are too rare; the factorization is what makes word-level viable there.

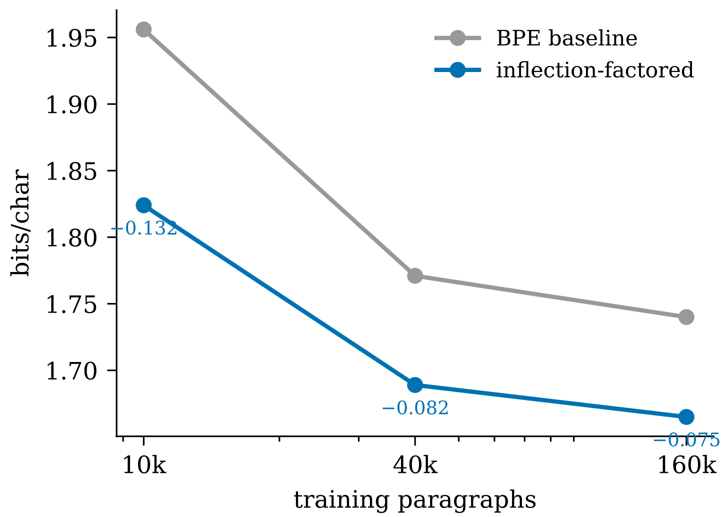

Figure 3: English data-efficiency ladder: the advantage over BPE widens as the corpus shrinks.

6 Measurement traps (perhaps the real contribution)

Nearly every strong effect was, on first measurement, an artifact favoring our hypothesis.

The `<unk>` cheat. BPE mapped out-of-vocab pieces to one frequent `<unk>` while bits/char credited the swallowed word's characters. BPE `<unk>`'d 5.7% of tokens, the lexeme stream 2.9%. Hence the "20% length penalty" was mostly BPE's cheat (true ratio 1.02×), and "derivation hurts" was an artifact: the more complete dictionary escaped less, cheated less, scored worse. A byte fallback fixed both.

Gauge freedom (four times). A free linear map in front of a structured operator makes its parameters unidentifiable; a randomly-initialized reflection once scored identically to a trained one. Pin the identity slot.

No-op nulls. Shuffling the *targets* of a fitted translation is permutation-invariant, so the "null" scored exactly what the fit scored. Permute the source or the root assignment.

Confounded arms. Changing dictionary, escape brackets and operator order in one run, then crediting whichever had a story. Minimal one-variable contrasts must come first.

Strawman baselines. Comparing a prototype init (std 0.02) against PyTorch's default $\mathcal{N}(0, 1)$; then leaving *positional* embeddings at $\mathcal{N}(0, 1)$ so word identity was 50× quieter than position.

Protocol dependence. The lexeme advantage moves with the LR schedule (−0.131 vs. −0.075 on identical configs). Any bits/char→params conversion must fix one protocol.

A related negative result. Initializing the LM's embedding from BERT contextual prototypes performed *identically to a shuffled permutation* of them; freezing them cost 0.8 bits/char. A good map of a masked LM's representation space is not a good input parameterization for a causal LM.

7 Limitations

Single machine; 2–3 seeds; treebank-scale corpora (17k–518k tokens) for the cross-linguistic result, a 200k-paragraph wikitext subset for English. **Corpus size is not controlled across languages**: the treebanks range 17k–518k tokens, and since the gain is sample-complexity-driven (Finding 1a), the *affine–free* column conflates morphology with corpus size—a matched-token cross-linguistic sweep is the needed follow-up. Korean (no FEATS), Arabic (lemmatization artifact), and Czech (parse failure) were dropped; the sample is Indo-European-heavy plus Turkish/Finnish. No comparison to a modern large-vocab BPE at scale or to subword regularization; the trunk is tiny.

8 Takeaway

For a causal language model: **lemmatize and tag inflection.** It is near zero-cost (1.02× tokens with a byte fallback, zero inference FLOPs), it helps most exactly where data is scarce and morphology is rich, and it is a structured-pooling effect more than a linguistic-correctness one. Derivation and the elaborate operator "language" we started with are not worth the complexity for an LM, though they remain valid descriptions of the geometry of English word embeddings.

A A sense-tokenization companion experiment

A separate exploration asked the dual question for *polysemy*: if morphology makes a lemma's meaning context-free by pooling its inflected forms, does splitting a word by *sense* help? We built a coarse sense inventory for 23,818 words (2.35 senses/word, including proper-noun and corpus-specific senses WordNet lacks) with a large model, then directly tagged 3.26M wikitext occurrences by sense via a batched API run (∼15× the size of SemCor). The dataset and dictionary are the deliverable; on top of them we trained two disambiguators and compared them on a common held-out set (Table 3).

Table 3: Sense disambiguators on a common 30k held-out set (*balanced* = word whose top sense is <75% of its occurrences). The trained cross-encoder wins everywhere once the evaluation is fair; a skew-routed ensemble is worse than the cross-encoder alone.

disambiguator	all	balanced	skewed
nearest-centroid propagator (frozen encoder)	0.743	—	—
cross-encoder (target-token readout + target-mask, trained)	**0.843**	0.740	0.896
skew-routed ensemble	0.779	0.783	0.773

Two lessons transfer from the main paper. First, *measure fairly*: the propagator appeared to win (0.879) under a per-word evaluation that only scored well-populated words; on a common all-words set it scores 0.743 and the cross-encoder wins cleanly, so the ensemble built on the propagator's apparent skewed-word strength loses. Second, *point the model where the signal is*: reading the target word's own contextual token rather than the sequence `[CLS]`, and injecting a target-mask embedding (segment-embedding style, zero-initialized) rather than bracketing the target in text, took balanced-word accuracy from 0.68 to 0.74—the earlier "0.68 ceiling" was undertraining and a mis-placed readout, not a limit. A *cased* backbone did not help: context already carries the disambiguating signal.